Exercise sheet 3

PRISM - Curves and Surfaces

1. Remember that to findt the arc length between two points p_1 and p_1, we choose a parametrization θ, where that $\theta(t_1) = p_1$ and $\theta(t_2)$ for some t_1 and t_2 and use this formula:

$$\int_{t_0}^{t_1} \|\theta'(t)\| dt$$

Why does this not depend on the parametrization?

2. These steps will show that the line segment between two points is the shortest possible curve between them.

   (a) Show that for any two vectors $\mathbf{v}$ and $\mathbf{w}$, the dot product $\mathbf{v}.\mathbf{w} \leq \|\mathbf{v}\|\|\mathbf{w}\|$.

   (b) Show that $\|\mathbf{v}\| = \mathbf{v}.\frac{\mathbf{v}}{\|\mathbf{v}\|}$. In other words, we can figure out the magnitude of a vector by taking the dot product with a unit vector in the same direction. Apply this to $\|\theta(t_1) - \theta(t_0)\|$ and the fundamental theorem of calculus and the previous parts to show that this distance between the two points is smaller than the arc length of θ between $\theta(t_0)$ and $\theta(t_2)$.

3. Note that the vectors $\mathbf{T}(t)$ and $\mathbf{N}(t)$ at the point $\theta(t)$ gives us a system of coordinates for all vectors originating at $\theta(t)$. What are the vectors $\mathbf{T}'(t)$, $\mathbf{N}'(t)$, $\mathbf{T}''(t)$, and $\mathbf{N}''(t)$ in terms of $\mathbf{T}(t)$ and $\mathbf{N}(t)$? For example, the vector $\mathbf{T}''(t) = x(t)\mathbf{T}(t) + y(t)\mathbf{N}(t)$ for some $x(t)$ and $y(t)$. What are those coefficients, $x(t)$ and $y(t)$? (*Hint:* We did similar things during the lecture. We used the fact that $\mathbf{T}(t)$ and $\mathbf{N}(t)$ are perpendicular, i.e. their dot product is 0.

4. Use the previous question to answer this: Given a curve parametrized by θ, we can define a new curve called the parallel of the original curve by parametrizing it by $\theta_c(t) = \theta(t) + c\mathbf{N}(t)$. (Remember that $\mathbf{N}(t)$ is that vector at $\theta(t)$ which is obtained by turning the unit tangent, $T(t)$, clockwise by $\pi/2$). What is the parallel of a circle of radius r? In general (not just for circles), what is the curvature of θ_c at the point $\theta_c(t)$ in terms of the curvature of θ at the point $\theta(t)$? What about the velocity vectors?

5. Supposing I give you a smooth function $\kappa(t)$, can you always find a curve parametrized by some parametrization θ so that its curvature at $\theta(t)$ is $\kappa(t)$?

6. If θ is a spatial curve then we cannot define $\mathbf{N}(t)$ by rotating $\mathbf{T}(t)$ in the clockwise direction because there is no natural plane to rotate it in. Instead, we could define $\mathbf{N}(t)$ as the unit vector in the direction of the acceleration. In that case, we have only two vectors in 3 dimensional space, so it is not enough to act as a basis in which to write the other vectors in terms of. But we can get a third one by taking the cross product $\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t)$. Once again, we should be able to write any vector at $\theta(t)$ in terms of $\mathbf{T}(t)$, $\mathbf{N}(t)$, and $\mathbf{B}(t)$. Show that $B'(t)$ must be in the direction of $\mathbf{N}(t)$ and that if we define $\tau(t)$ to be such that $\mathbf{B}'(t) = \tau(t)\mathbf{N}(t)$ and $\mathbf{T}'(t) = \kappa(t)\mathbf{N}(t)$, then we use $\kappa(t)$ and $\tau(t)$ to write all the derivatives of $\mathbf{T}$, $\mathbf{N}(t)$, and $\mathbf{B}(t)$ in terms of $\mathbf{T}$, $\mathbf{N}(t)$, and $\mathbf{B}(t)$.